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Conserved geometry of cell-type identity: a mammalian configuration and a replicating two-layer structure in primates

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Abstract
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The relative positions of cell types in transcriptomic space — the geometry of cell-type identity — are widely assumed to be conserved across species, but which features of that geometry are conserved, and which can be measured reliably in current data, have not been separated. We represent each cell type as a landmark, its mean expression across orthologous genes, and use Procrustes superimposition to decompose the cross-species geometry into two levels: the configuration of cell types relative to one another (Layer 1), and the orientation of each type's internal expression spread (Layer 2). In human and mouse, the configuration is conserved well beyond chance (observed-to-null ratio 0.52; $p < 10^{-6}$), carried disproportionately by the most evolutionarily conserved genes and enriched for canonical identity transcription factors. The within-type covariance orientation is also conserved, and — on three independent, same-protocol primate atlas pairs — the decomposition replicates: the configuration-optimal alignment compresses the covariance axis, and under gene-level standardization that axis resolves to cell-identity markers (all $p \le 10^{-4}$). Per-type divergence rankings behave differently. They are precise within an atlas but do not reproduce across atlases, and a simulation shows why: at the signal strength of real data, rank recovery saturates at a ceiling ($\rho \approx 0.42$) that additional sequencing cannot lift. Between-atlas technical variation is comparable to between-species biological variation at this resolution. Conserved cell-type geometry is measurable and replicates; per-type divergence is not yet resolvable in available atlases.

Keywords: Procrustes analysis, cross-species transcriptomics, cell type evolution, single-cell RNA-seq, covariance conservation, cell type geometry

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Author Summary
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Every cell type carries a characteristic pattern of active genes that places it at a particular position relative to all the others, and together those positions form an arrangement with a definite shape. That arrangement is widely assumed to be shared across animals, but it is rarely measured directly, and two questions about it are easy to run together: is it conserved across species, and can today's data actually measure its finer features? To ask both, we borrowed a method from comparative anatomy — the geometry used to compare skulls by aligning matching landmarks — and applied it to the average expression profile of each matched cell type in human, mouse, mouse lemur, macaque, and marmoset. The arrangement is conserved, measurably more than chance, and is carried by the most evolutionarily conserved genes, which include the master regulators of cell identity. Its two-layer structure — where each type sits, and how each type's cells spread around it — is conserved as well and replicates on independent primate data. But the popular per-type readout, the ranking of which cell types have "diverged most," cannot be recovered reliably from the atlases we have now: two atlases of the same species can disagree as much as two species do. That boundary matters for how such rankings should be read.

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INTRODUCTION
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Cell-type geometry, and two questions about it.

A cell type occupies a characteristic position in transcriptomic space, set by which genes it expresses and how strongly. Taken together, the cell types of an organism form an arrangement — some types close, some far, the whole with a definite shape. This arrangement is what we mean by the geometry of cell-type identity, and whether it is preserved across species is a natural way to ask whether cell-type identity itself is evolutionarily conserved.

To make the arrangement comparable across species, we borrow the framework geometric morphometrics uses to compare anatomical forms[12]. There, a form is reduced to a configuration of landmarks — corresponding points, like the same anatomical feature on two skulls — and two forms are compared by superimposing their landmarks to remove differences in position, orientation, and overall scale. What remains after superimposition is shape. We treat cell types as landmarks: each type's mean expression vector is a point, all types from both species are placed in a shared space of orthologous genes, and Procrustes superimposition aligns the two species' configurations. What remains is the conserved shape of the cell-type arrangement.

That arrangement has two levels, and we keep them separate throughout because they turn out to behave differently. Layer 1 is the configuration: where the cell-type centroids sit relative to one another. Layer 2 is the dispersion: the orientation of the cloud of cells around each centroid. For each layer we ask two questions that are easy to conflate. First, is the structure conserved between species? Second, is it recoverable from single-cell atlases as they exist today — is what we measure a property of biology, or of the particular atlases we measured? These are different questions, they have different answers here, and separating them is the point of this work: the conserved arrangement is real and reproducible, while one widely reported quantity — the ranking of which cell types have diverged most — turns out not to be recoverable from current data at all.

Comparative single-cell studies have developed a rich toolkit for relating cell types across species, but it answers a different question from the one above. Methods such as SAMap[3], SATURN[4], CAMEX[7], and optimal-transport[8] and foundation-model[10] approaches optimize cell-level matching — they establish which cell type in one species corresponds to which in another, producing correspondence scores, integrated embeddings, or transport probabilities. These are answers to "which maps to which." None produces a measure of how much a given cell type's position has shifted under a shared geometric transform — a per-type quantity that treats the cross-species comparison as a single alignment of two configurations and reads off what each type contributes to the residual. That per-type displacement, and the conservation of the overall configuration it presupposes, is what a geometric-morphometric treatment supplies and what existing correspondence methods do not.

The geometry itself has been shown to carry biological information within a species. Bulk transcriptomes of normal cell types exhibit a tree-like organization that is degraded in cancer[1], establishing that the arrangement of cell types — not only their individual expression profiles — reflects real biology. Whether that arrangement is conserved across species, and whether its conservation can be decomposed into separable geometric components, has not been quantified with single-cell resolution, where individual cell types can be dissociated from tissue composition.

Here we address both the conservation question and the measurement question together, and find they have different answers. Applying Procrustes superimposition to ontology-matched cell-type centroids, we show that the configuration of cell types is conserved across mammalian species; that a second geometric layer, the orientation of within-type covariance, is conserved as well and — on independent, same-protocol primate atlases — replicates; and that these conserved features are carried by evolutionarily conserved cell-identity genes. We then show that the per-type divergence ranking the same framework produces, though widely reported and precise within any single atlas, is not recoverable across atlases at the sequencing depths now available. Conserved cell-type geometry is measurable and reproducible; the finer question of which cell types have diverged most is, for current data, not yet resolvable — a boundary that bears directly on how cross-species divergence estimates from single atlases should be interpreted.

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RESULTS
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1. The configuration of cell types is conserved across species

Across 35 cell types matched between human and mouse by Cell Ontology label, the arrangement of cell-type centroids in transcriptomic space is conserved far beyond chance. Representing each type as its mean expression across 16,959 one-to-one orthologs and aligning the two species' configurations by Procrustes superimposition, the observed cross-species disparity is roughly half that of randomized configurations (observed-to-null ratio 0.52; $p < 10^{-6}$, one million permutations; Fig 1A,B). Lower ratios indicate stronger conservation, and 0.52 places the cell-type arrangement well inside the conserved regime.

Two controls establish that this reflects the biology of the arrangement rather than the way it was measured. The signal is not manufactured by the shared embedding: when each species is embedded independently and the configurations aligned without any joint basis, conservation is at least as strong (S1 Fig). And it is not merely lineage-level clustering — immune cells near immune cells, epithelial near epithelial — because a permutation test that shuffles labels only within lineage blocks, holding coarse structure fixed, still finds the observed configuration significantly ordered (obs/null 0.67; $p < 10^{-4}$; Fig 1C). Cell types occupy conserved positions relative to their neighbors, not just conserved neighborhoods.

The conservation is not confined to a single atlas pair, and the mammalian reach of the result rests on the human–mouse comparison, which spans the deep primate–rodent divergence (~90 million years[15]). A second, independently generated pair — human and mouse lemur, a strepsirrhine primate separated from human by roughly 75 million years[15] — clears the same test (obs/null 0.35; $p < 10^{-4}$; Fig 1D), an independent replication of the configuration signal in a different primate lineage rather than at a deeper divergence. Conserved cell-type configuration is thus a property of more than one species pair: established across mammals by the primate–rodent comparison and reproduced, independently, within primates.

A configuration, though, is only the coarsest description of the geometry. It records where each cell type sits, but not the shape of the variation around it. Whether conservation reaches that finer structure is the next question.

2. Both geometric layers are conserved, and the decomposition replicates on independent atlases

The arrangement has a second level. Beyond where each centroid sits (the configuration, Layer 1) is the orientation of the cloud of cells around it — the covariance ellipsoid of each cell type (Layer 2). These are conserved as distinct properties: the alignment that optimally superimposes the centroids does not also absorb the covariance orientation but compresses it, leaving a significant residual (Krzanowski[23] subspace similarity dropping under the centroid-optimal rotation but remaining above its null, $p < 10^{-4}$; Fig 2A,B). Position and dispersion carry separately conserved information.

Because the covariance is estimated from individual cells, it invites a question the configuration does not: is the conserved orientation a property of the biology, or of the specific atlases measured? We tested this directly, on three cross-species atlas pairs generated under a single protocol — human, macaque, and marmoset basal ganglia from a harmonized 10x atlas, 52 to 55 matched cell types, with a few hundred to over 100,000 cells per type. In all three pairs the two-layer decomposition replicates (Fig 2C): the configuration-optimal alignment compresses the covariance axis in every pair, at every subspace dimension tested, each beyond its permutation null (all $p \leq 10^{-4}$). And the conserved axis is biological, not technical. Under per-gene standardization — which removes the dominance of a few high-variance genes — the leading conserved covariance axis resolves to canonical cell-identity markers rather than to housekeeping structure.

One cross-species comparison does not settle this question, and it is worth saying why. Comparing whole-cell to single-nucleus data (a protocol switch, rather than an independent same-protocol pair) confounds biology with assay: nuclear and cytoplasmic RNA sample different molecular pools, and in nuclear data the leading covariance axis is dominated by transcript length rather than cell identity (S2 Text). The same-protocol primate replication is the informative test, and it supports a shared, identity-bearing covariance axis conserved across species.

The geometry, then, is conserved at both levels and reproduces on independent data. How widely the configuration result itself holds — across the many atlases now available, and across greater evolutionary distances — is the next question.

3. The configuration result is robust across atlases and divergence times

The configuration signal is not specific to one atlas pair. Across independent replication datasets it recurs at comparable strength (obs/null 0.45–0.55, each $p < 10^{-4}$; Fig 3), including a fully independent pair that shares neither of the original atlases. Where the signal weakens, the reason is identifiable rather than biological: one non-replication is simply underpowered, at six matched types, with an effect size consistent with the others; a second traces to a microwell-based protocol, and a within-human control on the same data localizes the failure to that protocol while itself remaining coherent (S2 Fig). The conserved configuration is a property of the biology that survives changes of atlas, not an artifact of any one dataset.

A human–macaque comparison is consistent in direction but does not clear multiple-testing correction on the twelve available cell types, and we report it without interpreting ratios across species pairs (S1 Text).

The global configuration, then, is robust wherever it is tested. This makes the behavior of the per-type readout derived from the same alignment — the ranking of which cell types have diverged most — all the more striking, because it does not share this robustness at all.

4. Per-type divergence rankings are precise within an atlas but not resolvable across atlases

The same alignment that gives the conserved configuration also produces a per-type readout — a divergence ranking, ordering cell types by how far each has moved between species — that is widely reported and behaves nothing like the global signal. Within a single atlas it is highly reproducible: bootstrap test–retest correlation reaches $\rho = 0.99$, and every cell type's rank is tightly bounded (Fig 4A). Across atlases it collapses. The same cell types re-order almost arbitrarily between atlas pairs (cross-atlas $\rho \approx 0.15$), and — revealingly — the types ranked most precisely within an atlas are the ones that move most across atlases (Fig 4B). Precision within a dataset and reproducibility across datasets are not merely different here; they are inversely related, which is the signature of a measurement dominated by dataset-specific structure.

A simulation identifies the ceiling. Planting a known divergence ranking, adding noise calibrated to the real data, and attempting recovery, the correlation between the true ranking and the recovered one saturates at $\rho \approx 0.42$ at the signal strength of the actual atlases — and it saturates there regardless of how many cells are sampled (Fig 4C). The cross-atlas instability is therefore not sampling noise, and it is not a problem that deeper sequencing can fix: the information required to resolve per-type divergence is not present in the data at this level, at any sequencing depth.

This has a concrete consequence for how such rankings should be read. At the level of cell-type geometry, between-atlas technical variation is comparable to between-species biological variation — two atlases of the same species can differ as much as two species do (S7 Fig). A per-type divergence ranking computed from a single atlas pair is therefore an atlas-specific observation, not a resolved biological ordering, and distinguishing genuine divergence from atlas composition will require harmonized multi-atlas designs rather than deeper sequencing of existing ones.

Consistent with divergence being unresolved rather than driven by any single known factor, ten pre-specified mechanistic explanations for the per-type signal — cell count, sequencing depth, network centrality, sequence-conservation covariates, and others — were tested, and none reached significance at this sample size, ruling out strong but not moderate associations (S1 Table, S1 Text). A separate control bears on a related question: whether cross-species divergence exceeds ordinary individual variation. Splitting donors within an atlas, cross-species divergence is directionally larger than within-individual variation across all resamples, though this contrast is confounded with atlas and protocol (the cross-species arm spans two atlases, the within-species arm one) and so establishes direction, not a species-level effect size.

5. The conserved configuration is carried by evolutionarily conserved identity genes

Finally, the conserved arrangement is carried by the genes evolution has most conserved. Restricting the analysis to the top quartile of genes by cross-species conservation preserves the configuration signal, and these genes are enriched for canonical master and identity transcription factors against backgrounds matched for both expression level and cell-type specificity (median conservation percentile 0.94, versus 0.54 for an expression-matched background and 0.76 for a background matched jointly on expression and specificity; Fig 5). The conserved cell-type arrangement is thus identified with conserved cell-identity regulatory programs — a quantification of what the conserved geometry represents, rather than a new claim about regulation.

Taken together, the results draw a clean line through what current data can and cannot support. The configuration of cell types is conserved across mammals, replicates at the level of within-type covariance on independent primate atlases, and is carried by conserved identity genes. The per-type divergence ranking that the same framework produces — and that is widely reported — is precise within an atlas but not recoverable across atlases at any sequencing depth now available. Conserved cell-type geometry is measurable; the finer question of which types have diverged most is, for now, beyond the reach of the atlases we have.

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DISCUSSION

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Treating cell types as landmarks and comparing species by superimposition separates two things that a single similarity score conflates: whether the arrangement of cell types is conserved, and whether the shape of variation around each type is conserved. Both are, and they are conserved as distinct properties — the alignment that best matches the centroids compresses, rather than absorbs, the covariance orientation, so position and dispersion carry separable conserved information. That this two-layer structure replicates across three independent primate atlas pairs generated under a single protocol, and that its conserved covariance axis resolves to cell-identity markers under gene-level standardization, indicates a shared biological organization rather than a statistical coincidence of one dataset.

The more consequential result is a boundary. The per-type divergence ranking — which cell types have diverged most between species — is among the most natural quantities to extract from a cross-species comparison, and it is widely reported. We find it is precise within any single atlas but not reproducible across atlases, and a simulation shows why: at the signal strength of real data, the recoverable correlation between a true divergence ranking and the estimated one saturates at $\rho \approx 0.42$ regardless of sequencing depth. The instability is a property of the information available in current atlases, not of sampling. In practical terms, between-atlas technical variation is comparable to between-species biological variation at this level of geometry, and a divergence ranking from a single atlas pair should be read as atlas-specific rather than as a resolved biological ordering. This does not mean cross-species divergence is unmeasurable in principle; it means that measuring it will require harmonized, multi-atlas designs that control annotation and composition across sources — a design constraint, not a deeper sequencer.

Three bounds delimit these conclusions. First, the covariance-layer replication is demonstrated across primates; whether it extends to greater divergence, such as the primate–rodent split at which the configuration is conserved, remains to be tested with same-protocol atlases at that distance. Second, the divergence-ranking result is a statement about currently available atlases, established on the species and tissues examined here; it identifies a resolution limit under present data rather than a permanent one. Third, the framework compares configurations of ontology-matched types and inherits the granularity of those annotations — a coarsely annotated type and a finely annotated one are not equivalent landmarks, which is one reason harmonized annotation is the operative requirement for resolving per-type divergence. Each of these is a boundary on scope, and none is contradicted by the results within it.

The conserved configuration is carried disproportionately by the most cross-species-conserved genes, which are enriched for master and identity transcription factors. We read this as a quantification of the identity interpretation — the conserved arrangement is the arrangement of cell identities, encoded by conserved regulatory genes — rather than as a discovery about regulation itself; the enriched genes are, in part, conserved by the same evolutionary pressures that conserve the geometry, and we do not separate those. The value of the result is that it ties an abstract geometric signal to a concrete and independently sensible molecular basis.

The two-layer decomposition is a general instrument: any pair of cell-type collections in a shared feature space can be compared this way, and the same machinery that here separates conserved configuration from conserved dispersion could compare configurations across conditions, developmental stages, or perturbations. The immediate methodological contribution, though, is the pair of findings taken together — that conserved cell-type geometry is real, replicable, and identity-bearing, and that the per-type divergence structure the same method produces is not yet resolvable in available data. Reporting both, rather than the positive alone, is what the measurement supports; it marks the line between what current single-cell atlases can establish about cell-type evolution and what they cannot yet.

Materials and Methods

We refer to the framework used throughout — landmark-based Procrustes superimposition of ontology-matched cell-type centroids, with a two-layer decomposition into configuration and within-type covariance — as CellWarp. All analyses were implemented in Python 3.12 with a global random seed of 42; the version-pinned dependency manifest, the archived ortholog tables, and all scripts are deposited with the code (Data and code availability). Dated internal analysis plans were committed to the repository before the corresponding statistics were computed; "pre-specified" throughout denotes a design or threshold decision fixed before the corresponding result was observed. No analysis was registered with an external public pre-registration registry; some of the deposited plans cover analyses reported elsewhere or held in reserve rather than in this paper, and the full classification of every script, analysis, and plan (paper, banked, other-project, exploratory) is provided in the repository as SCOPE.md.

Data and cell-type matching

Primary human data were from Tabula Sapiens[25] (v1.0; 483,152 cells, 24 donors, 10x Chromium and Smart-seq2) and primary mouse data from Tabula Muris Senis[26] (whole-organism, multiple ages, 10x Chromium and Smart-seq2), both accessed via CZ CELLxGENE Census[29] (version 2025-11-08), which is the reproducibility anchor for these datasets. The mouse-lemur extension used Tabula Microcebus[19] (Microcebus murinus, 10x 3′ v2, accessed via CZ CELLxGENE Discover). The primate covariance replication (Layer 2) used the harmonized 10x basal-ganglia atlas of the Human and Mammalian Brain Atlas consortium (human, macaque, and marmoset). Because both atlases integrate across multiple tissues per organism, each cell-type centroid aggregates across tissue contexts rather than reflecting a single organ; because Tabula Muris Senis spans several ages whereas the human atlas is predominantly adult, an age-composition difference between the two species is not controlled and is a potential contributor to per-type residuals.

Cross-species correspondence was defined operationally by shared Cell Ontology[41] label: a type was included when it appeared under an identical canonical Cell Ontology display string in both atlases and reached at least 500 cells per species after quality filtering. No marker re-annotation was applied. This treats a shared label as a correspondence, not an assertion of evolutionary homology[17]; for broad parent labels (for example stromal cell, epithelial cell, generic monocyte) it may aggregate subtypes whose cross-species correspondence is not established, so residuals at those labels confound divergence with cross-species subtype-composition differences that the framework does not separate. Under the Census filters, 66 labels were shared across the two atlases and 35 met the 500-cell threshold, spanning immune/hematopoietic, epithelial, stromal/mesenchymal, endothelial, and metabolic/parenchymal lineages (full list, S5 Table). A 6-type subset (B cell, CD4+ T cell, CD8+ T cell, endothelial cell, hepatocyte, macrophage) is used where a smaller matched n is required.

Human–mouse one-to-one orthologs were obtained from Ensembl BioMart release 115 (GRCh38.p14 / GRCm39; accessed 2026-03-12); only strict 1:1 relationships were retained (paralog averaging was not performed, which deflates rather than inflates any conservation signal). After intersecting with genes present in both atlases, the operating gene space comprised 16,959 orthologs. The full ortholog table is archived as CSV and is the reproducibility anchor for all gene-space definitions. A maximum of 2,000 cells per type per species was randomly subsampled (seed 42) before centroid computation for the primary human–mouse analysis; no per-type subsampling was applied for the covariance (Layer 2) analyses, which use all available cells of each type.

The Procrustes framework

Cell-type centroids were computed as the mean of log1p-transformed, library-size-normalized (counts per 10,000; CP10K) expression over all cells of a type within a species, independently per species with no cross-species alignment, integration, or batch correction; the cross-species signal is therefore present in the raw centroid displacement, not introduced by alignment. Because centroids are means of log1p expression, high-mean ubiquitous genes (ribosomal, housekeeping) contribute disproportionately to centroid and within-type-covariance geometry — the reason the leading unscaled within-type axis is ribosomal-dominated (below).

Prior to superimposition, principal component analysis was applied jointly to the stacked matrix of 70 centroids (35 types × 2 species) in the 16,959-gene space, retaining the number of components reaching 95% cumulative variance (33 components, 95.2%). Dimensionality reduction is required for numerical stability: with only 70 landmark points in thousands of dimensions, Procrustes alignment is otherwise underdetermined. Rankings were stable across $k = 10$–50 (S2 Fig). Procrustes superimposition was implemented directly (not via a library routine) to expose per-type residuals in the original gene space. Each configuration was centered; the cross-covariance $M = Xc^{\top}Yc$ of the centered human (Xc) and mouse (Yc) configurations was decomposed by singular value decomposition $M = U\Sigma V^{\top}$; the rotation $R = VDU^{\top}$ (D diagonal with final entry $\mathrm{sign}(\det(VU^{\top}))$, others 1) algebraically guarantees $\det(R) = +1$, precluding reflections; and the rotated mouse configuration was scaled by the optimal full-Procrustes scalar. Procrustes distance is the sum of squared residual displacements; per-type residuals are the Euclidean distances between aligned human–mouse centroid pairs (higher = more divergent). The custom implementation was verified against scipy.spatial.procrustes to machine precision.

Significance was assessed by a permutation null that shuffles cell-type labels (not cells within types) in the mouse configuration, testing whether the global transform aligns all types simultaneously better than chance; the empirical p-value uses the conservative (n ≤ + 1)/(nperm + 1) estimator. The primary human–mouse test used 1,000,000 permutations (observed distance 61.15; nearest null 98.88; zero permutations ≤ observed; $p < 10^{-6}$); replications, extensions, and secondary tests used 10,000 permutations (floor $p < 10^{-4}$). The observed-to-null ratio (observed distance / median permuted distance) is the effect size; because the null median varies with type count, obs/null is not comparable across analyses with different n. A lineage-stratified null (labels shuffled only within five lineage blocks) tests positional correspondence beyond lineage membership (obs/null 0.668, $p < 10^{-4}$). Robustness was confirmed by bootstrap resampling (50% of cells per type; Procrustes-distance CV = 0.004 over 100 iterations) and by leave-one-out cross-validation with PCA and rotation refit per fold (35/35 held-out types predicted better than random pairing). An independent-PCA variant (per-species PCA aligned via loadings, label-free) and a Mantel test on pairwise-distance matrices are reported as artifact-ruling controls (S1 Fig; S1 Text).

The two-layer decomposition

Layer 1 is centroid position (the primary Procrustes result). For Layer 2, single cells of each type were projected into the same 33-component joint PCA space, the per-type 33×33 covariance was computed (all cells; eigendecomposition by NumPy linalg.eigh), and the leading k eigenvectors define each type's covariance-ellipsoid orientation. Cross-species alignment was quantified by the Krzanowski[23] subspace-similarity statistic $S(k) = \|A^T B\|_F^2 / k$ (A, B the k leading eigenvectors for the two species; range 0–1, higher = better alignment) at $k \in \{1, 3, 5\}$, against a 10,000-permutation label-shuffle null. Without constraint to the centroid alignment, Layer 2 was significant at k = 5 (S = 0.483 vs null 0.375; $p < 10^{-4}$) and k = 3, but not k = 1 (single-axis alignment is too narrow to detect). The centroid-optimal rotation R was then applied to each type's mouse-side covariance eigenvectors and re-aligned to the unrotated human eigenvectors: this compresses aggregate alignment from S = 0.483 to S = 0.230 at k = 5, and the compressed statistic remains above its (fixed-R) null (S = 0.230 vs null 0.180; $p < 10^{-4}$). Position and dispersion therefore carry distinct conserved information; the post-rotation test is conditional on the observed R, not a marginal test of Layer 2. The leading conserved axis per type (CPC1; top eigenvector of the cell-count-weighted covariance sum) was projected back to gene space and ranked by loading. Under the unscaled weighting, 25 of 35 types have a ribosomal-protein rank-1 driver (a weighting property of the unscaled covariance, not a housekeeping claim); under per-gene standardization this collapses to 1 of 35 and the axis resolves to cell-type markers (S9 Table). Excluding ribosomal-protein genes leaves the Layer 2 statistic significant (post-rotation S = 0.234 vs null 0.180; $p < 10^{-4}$). The load-bearing basis for treating Layer 2 as a conserved statistic rather than a single shared biological axis is the cross-protocol non-replication of its driver genes, established next.

Primate replication (basal ganglia)

The Layer 2 decomposition was replicated on the harmonized 10x basal-ganglia atlas across three same-protocol cross-species pairs (Human–Macaque, Human–Marmoset, Macaque–Marmoset; 52 to 55 matched cell types, per-type cell counts from a few hundred to over 100,000). Using a single 10x protocol across all three species removes the whole-cell-versus-single-nucleus assay confound that makes the human–mouse-to-PanSci protocol switch uninformative (S2 Text). For each pair, centroids and per-type covariances were computed on the shared ortholog space, joint PCA was fit per pair at the 95% variance threshold, and the two-layer statistics S(k) were computed pre- and post-centroid-rotation at $k \in \{1, 3, 5\}$ against a 10,000-permutation label-shuffle null, exactly as in the primary analysis. Two weightings were run: unscaled (Scheme W0) and per-gene standardized (Scheme B, W2), the latter serving as the biological-axis control by removing the dominance of a few high-variance genes. In every pair, at every k, the centroid-optimal alignment compressed the covariance axis while remaining above its permutation null (all pre- and post-rotation $p \approx 10^{-4}$). Under unscaled weighting the rank-1 CPC1 driver was never a canonical identity marker; under per-gene standardization canonical markers surfaced (strongest in Human–Macaque), consistent with the conserved axis being identity-bearing rather than a technical high-variance artifact. Each type's covariance is estimated from that type's own cells, so the aggregate is weighted toward well-sampled types; the great majority of matched types are well sampled (42/55, 37/52, and 39/52 types have at least 500 cells in both species, most in the thousands to over 100,000), and restricting the covariance aggregate to those types leaves the compression and its significance unchanged in all three pairs and both weightings (post-rotation still below pre-rotation at $k = 5$, all $p \approx 10^{-4}$; layer2_lowN_robustness.json). The result is therefore not an artifact of the sparsely-sampled minority of types. Per-pair statistics and CPC1 driver classifications are deposited with the basal-ganglia analysis (Data and code availability).

Simulation

Pipeline statistical properties were characterized by a plant-and-recover simulation. Synthetic paired human/mouse configurations were built by placing n cell-type centroids in a 50-factor latent space and embedding them into a 500-gene space via a scaled Gaussian projection; a Haar-distributed orthogonal rotation simulated the cross-species transform; and per-type Gaussian noise (log-normally distributed across types, ≈25× spread) planted a known per-type divergence ranking. A single signal-strength parameter controls the noise level; it was calibrated by grid search with interpolation to reproduce the observed primary obs/null of 0.522 (calibrated signal ≈ 3.68). Global detection power was 100% at the calibrated signal across n = 15, 25, 35, with a well-calibrated null (false-positive rate 4.8% at $\alpha = 0.05$). Per-type ranking recovery, measured as Spearman ρ between planted and recovered rankings, saturated at $\rho \approx 0.42$ at the calibrated signal and did not increase with cells per type (50–2,000). Within-atlas test–retest (two independent cell samples from the same true centroids) reached $\rho = 0.994$. The recovery ceiling and detection-power panels are in Fig 4C and S1 Fig; per-condition summaries in S2 Table.

Data and code availability

All analysis code, the archived Ensembl/BioMart ortholog tables, and the pinned dependency manifest are deposited at Zenodo under the MIT License (https://doi.org/10.5281/zenodo.20735612) and mirrored on GitHub (https://github.com/sriramdevadas/cellwarp). Original analysis outputs are deposited at Zenodo under CC0 (https://doi.org/10.5281/zenodo.20735640); intermediate files regenerate from the code and the accessions below. The self-contained basal-ganglia primate-replication deposit (independent reproduction of the Fig 2 two-layer decomposition from the source h5ad, with the CellWarp method pinned as a versioned dependency) is archived at Zenodo (DOI: [BG_DEPOSIT_DOI – pending mint]). Source atlases were accessed via CZ CELLxGENE Census[29] (version 2025-11-08): Tabula Sapiens[25] (figshare 10.6084/m9.figshare.14267219) and Tabula Muris Senis[26] (figshare 10.6084/m9.figshare.12654728); Tabula Microcebus[19] via CZ CELLxGENE Discover (downloaded 2026-04-05); replication accessions Sun et al.[24] (OMIX002605 / GSA CRA007207), PanSci[27] (GEO GSE247719), Andrews et al.[30] (GEO GSE240429), Qu et al.[18] (GEO GSE196792), CellHint[28] via Census. The 15-dataset pan-Census UUID manifest is archived with the code.

Use of generative AI

In preparing this work, the author used Claude (a large language model developed by Anthropic; models in the Claude 3.5–Opus family) to write and generate analysis code, to assist with data analysis, and to draft and edit manuscript text. The author evaluated validity by reading and reviewing all AI-generated code, running the accompanying automated test and validation suites (including reproduce/validate.py, which checks reported statistics against tracked output files), and reproducing every reported result against its source artifact before inclusion. AI assistance affected the analysis code, figure generation, and the wording of the text; it did not originate the study's conceptualization, design, hypotheses, interpretations, results, or conclusions, which are the author's own. All relevant sources are cited, and the author takes full responsibility for the accuracy, validity, and integrity of all code, analyses, results, and text. Claude is not, and cannot be, an author of this work.

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AUTHOR CONTRIBUTIONS
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Sriram Devadas: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing: original draft, Writing: review & editing, Visualization, Project administration.

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FIGURE LEGENDS

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Fig 1. The configuration of cell types is conserved across mammalian species. (A) Pipeline. Cell-type centroids are computed independently for human and mouse over 16,959 shared 1:1 orthologs, projected into a 33-component joint PCA space (95.2% variance), aligned by Procrustes superimposition, and read out as per-type residual displacements. (B) The observed cross-species Procrustes distance (61.15) falls far below the 1,000,000-permutation label-shuffle null (obs/null 0.52; $p < 10^{-6}$): the human–mouse configuration is conserved well beyond chance. (C) The conservation is not just lineage-level clustering — permuting labels only within lineage blocks still leaves the configuration significantly ordered (obs/null 0.67; $p < 10^{-4}$). (D) The signal holds across mammalian divergence: a human–mouse-lemur pair (~75 Mya) clears the same test (obs/null 0.35; $p < 10^{-4}$).

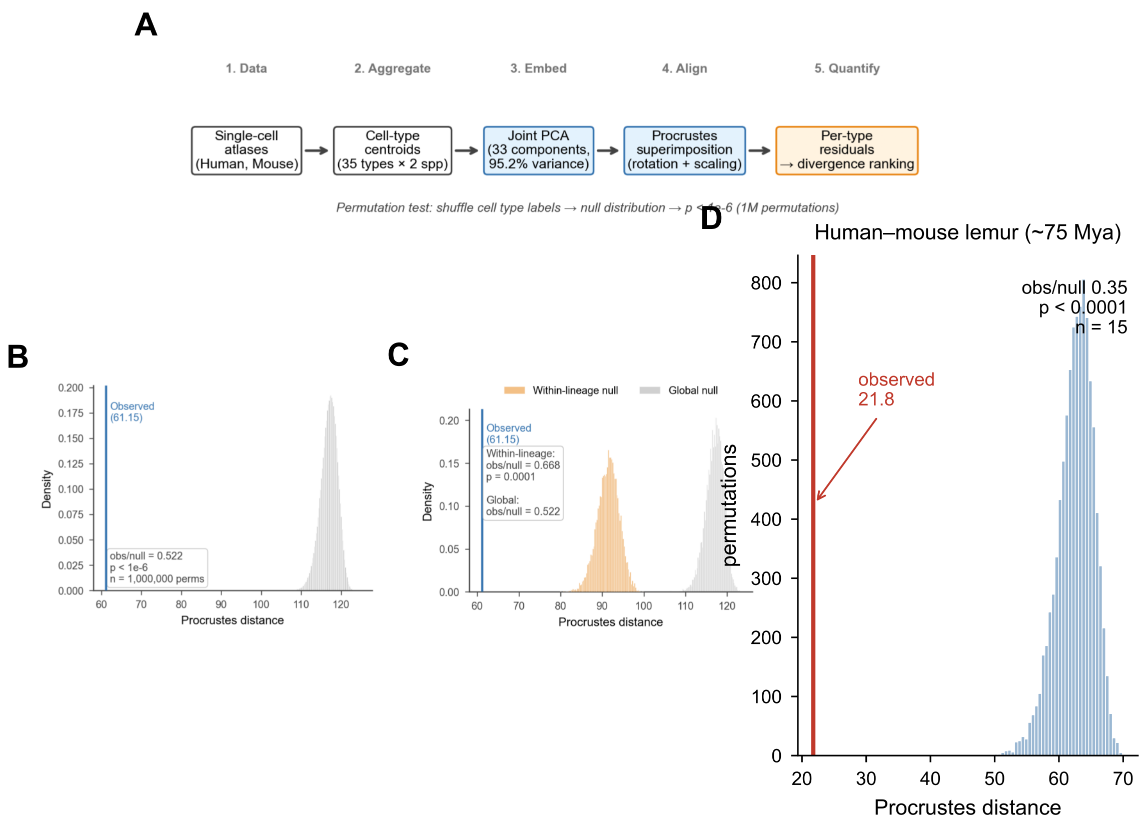
Fig 2. Both geometric layers are conserved, and the two-layer decomposition replicates on

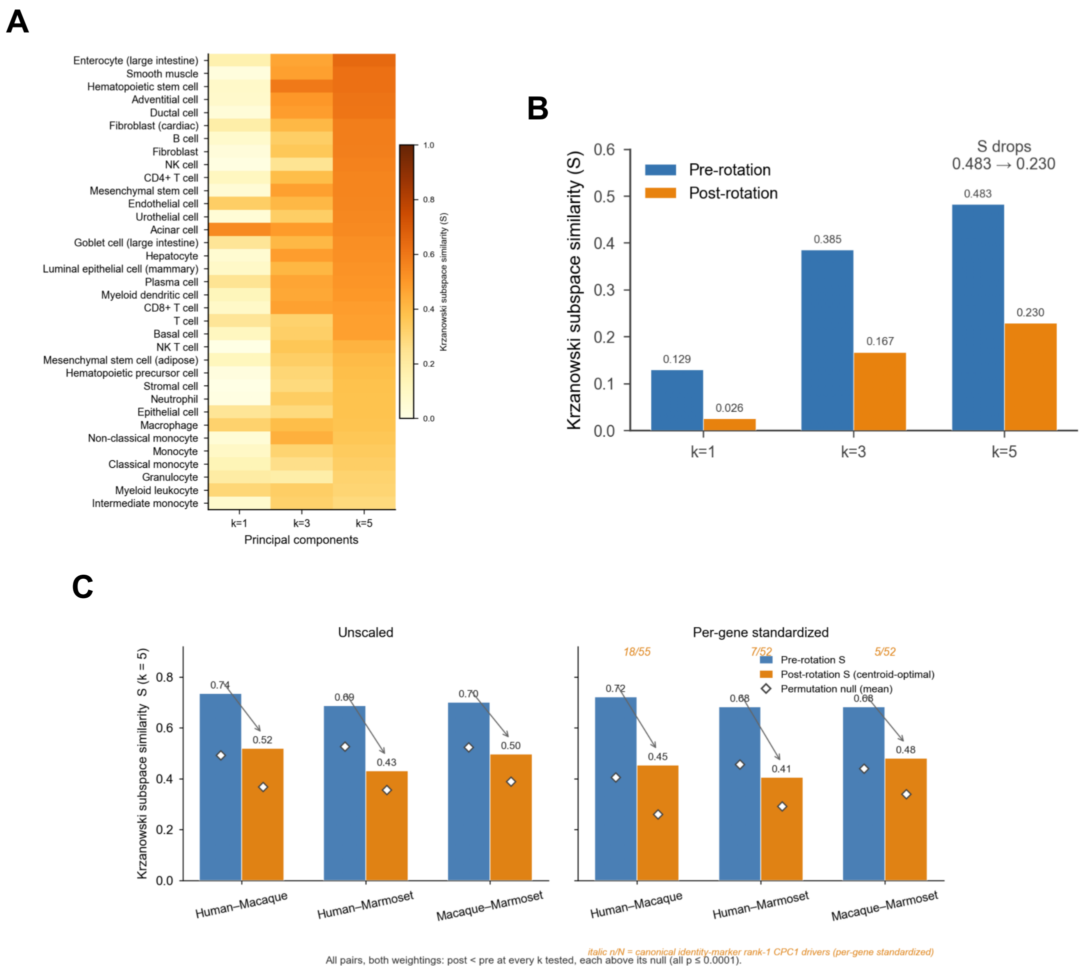
independent primate atlases.
(A) Position and dispersion carry separately conserved information: the within-type covariance-ellipsoid alignment (Krzanowski S) is significant against its label-shuffle null. (B) The centroid-optimal rotation compresses, rather than absorbs, the covariance alignment (S 0.483 → 0.230 at $k = 5$), and the compressed statistic stays above its null ($p < 10^{-4}$) — the two layers are geometrically distinct. (C) The decomposition replicates on three same-protocol primate pairs from a harmonized 10x basal-ganglia atlas (Human–Macaque, Human–Marmoset, Macaque–Marmoset): the configuration-optimal alignment compresses the covariance axis in every pair, at every subspace dimension, each beyond its permutation null (all $p \leq 10^{-4}$), under both unscaled and per-gene-standardized weightings; under standardization the conserved axis resolves to cell-identity markers.

Fig 3. The conserved configuration is robust across independent atlases.
The global configuration signal recurs at comparable strength across independent replication datasets (obs/null 0.45–0.55, each $p < 10^{-4}$), including a fully independent pair that shares neither original atlas. Where the signal weakens the cause is technical and identifiable (an underpowered six-type liver atlas; a microwell-based protocol localized by a within-human control), not biological — the configuration survives changes of atlas.

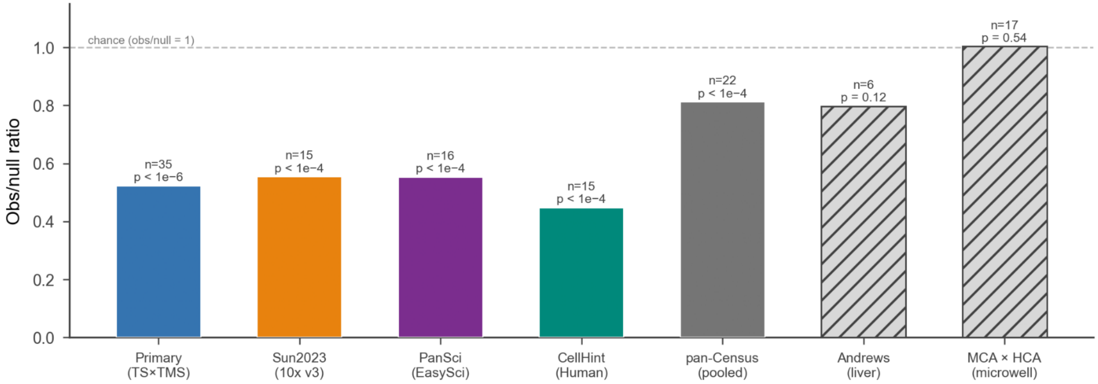
Fig 4. Per-type divergence rankings are precise within an atlas but not resolvable across atlases.
(A) Within a single atlas the per-type ranking is highly reproducible: bootstrap test–retest $\rho = 0.99$ and every cell type's rank is tightly bounded. (B) Across atlases it collapses (cross-atlas $\rho \approx 0.15$), and the types ranked most precisely within an atlas move most across atlases — precision and cross-atlas reproducibility are inversely related, the signature of a measurement dominated by dataset-specific structure. (C) A plant-and-recover simulation identifies the ceiling: at the signal strength of real data, recovery of the true ranking saturates at $\rho \approx 0.42$ regardless of how many cells are sampled — deeper sequencing cannot lift it.

Fig 5. The conserved configuration is carried by evolutionarily conserved identity genes.
Restricting to the top quartile of genes by cross-species conservation preserves the configuration signal, and those genes are enriched for canonical master and identity transcription factors against backgrounds matched for expression level and cell-type specificity (median conservation percentile 0.94, versus 0.54 expression-matched and 0.76 jointly matched). The conserved arrangement is thus identified with conserved cell-identity regulatory programs.

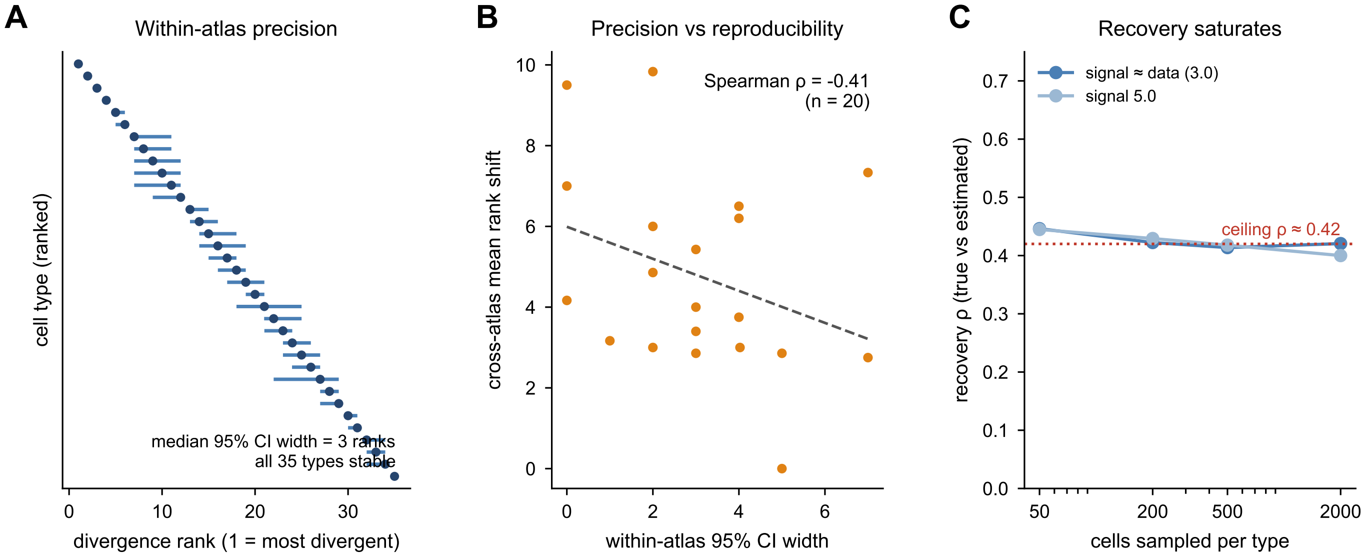

A
1. Data
2. Aggregate
3. Embed
4. Align
5. Quantify
Single-cell atlases (Human, Mouse)
Cell-type centroids (35 types × 2 spp)
Joint PCA (33 components, 95.2% variance)
Procrustes superimposition (rotation + scaling)
Per-type residuals → divergence ranking
Permutation test: shuffle cell type labels → null distribution → p < 1e-6 (1M permutations)
B
Observed (61.15)
obs/null = 0.522
p < 1e-6
n = 1,000,000 perms
Density
Procrustes distance
C
Within-lineage null
Global null
Observed (61.15)
Within-lineage:
obs/null = 0.668
p = 0.0001
Global:
obs/null = 0.522
Density
Procrustes distance
D
Human–mouse lemur (~75 Mya)
obs/null 0.35
p < 0.0001
n = 15
observed
21.8
permutations
Procrustes distance

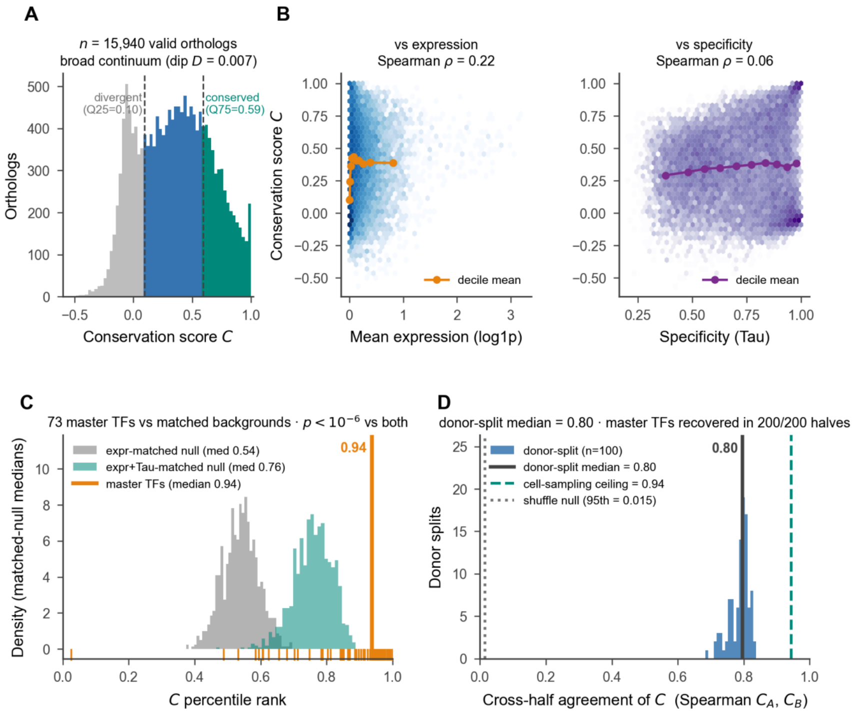


A
Enterocyte (large intestine)
Smooth muscle
Hematopoietic stem cell
Adventitial cell
Ductal cell
Fibroblast (cardiac)
B cell
Fibroblast
NK cell
CD4+ T cell
Mesenchymal stem cell
Endothelial cell
Urothelial cell
Acinar cell
Goblet cell (large intestine)
Hepatocyte
Luminal epithelial cell (mammary)
Plasma cell
Myeloid dendritic cell
CD8+ T cell
T cell
Basal cell
NK T cell
Mesenchymal stem cell (adipose)
Hematopoietic precursor cell
Stromal cell
Neutrophil
Epithelial cell
Macrophage
Non-classical monocyte
Monocyte
Classical monocyte
Granulocyte
Myeloid leukocyte
Intermediate monocyte
k=1
k=3
k=5
Principal components
Krzanowski subspace similarity (S)
1.0
0.8
0.6
0.4
0.2
0.0
B
Krzanowski subspace similarity (S)
Pre-rotation
Post-rotation
S drops
0.483 → 0.230
0.129
0.026
0.385
0.167
0.483
0.230
k=1
k=3
k=5
C
Unscaled
Per-gene standardized
Krzanowski subspace similarity S (k = 5)
18/55
7/52
5/52
Pre-rotation S
Post-rotation S (centroid-optimal)
Permutation null (mean)
0.74
0.52
0.69
0.43
0.70
0.50
0.72
0.45
0.68
0.41
0.68
0.48
Human–Macaque
Human–Marmoset
Macaque–Marmoset
italic n/N = canonical identity-marker rank-1 CPC1 drivers (per-gene standardized)
All pairs, both weightings: post < pre at every k tested, each above its null (all p ≤ 0.0001).



Obs/null ratio
0.0
0.2
0.4
0.6
0.8
1.0
chance (obs/null = 1)
n=35
p < 1e−6
n=15
p < 1e−4
n=16
p < 1e−4
n=15
p < 1e−4
n=22
p < 1e−4
n=6
p = 0.12
n=17
p = 0.54
Primary
(TS×TMS)
Sun2023
(10x v3)
PanSci
(EasySci)
CellHint
(Human)
pan-Census
(pooled)
Andrews
(liver)
MCA × HCA
(microwell)


A
Within-atlas precision
cell type (ranked)
median 95% CI width = 3 ranks
all 35 types stable
0
10
20
30
divergence rank (1 = most divergent)
B
Precision vs reproducibility
Spearman ρ = -0.41
(n = 20)
cross-atlas mean rank shift
0
2
4
6
8
10
within-atlas 95% CI width
C
Recovery saturates
signal ≈ data (3.0)
signal 5.0
ceiling ρ ≈ 0.42
recovery ρ (true vs estimated)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
50
200
500
2000
cells sampled per type



C = per-gene Pearson correlation of the expression profile across the 35 matched cell-type centroids, human vs mouse (computed independently of the Procrustes axes).